

# Quantum Computing for Fraud Detection

A credit-card fraud case study,  
the advantage curve, and the 24-month execution plan

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# Executive summary — the ask in one slide

## THE PROBLEM

Credit-card fraud cost Indian banking ~Rs 1,300 Cr in FY24 and is growing 20%+ YoY. Classical ML detects ~92-95% of fraud at the cost of high false-positives and slow adaptation to novel attack patterns.

## THE OPPORTUNITY

Quantum machine learning (VQC, quantum kernels) finds non-linear correlations classical models cannot reach — measurable gains in recall on rare-event fraud and the patterns we miss today.

## THE ASK

Rs 8-12 Cr over 24 months for a hybrid quantum-classical fraud platform: simulator-first proof on real HDFC data, then pilot on IBM Quantum hardware, then production tier-2 re-scorer.

## WHAT GOOD LOOKS LIKE IN 24 MONTHS

**+15-25%**

incremental fraud caught at the same false-positive rate

**– 30%**

reduction in customer-facing false declines on premium cards

**Rs 60-90 Cr 3 use cases**

annual loss avoidance at HDFC scale (conservative)

live in production — cards, UPI, and AML transaction monitoring

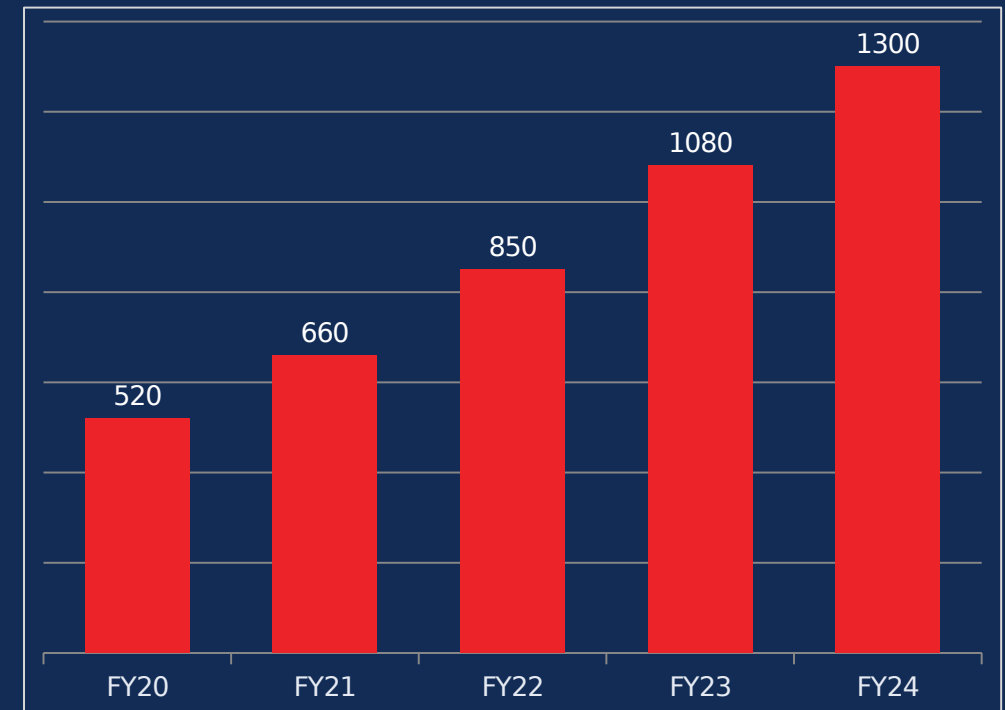
# The fraud problem at HDFC scale

## Why this matters at the bottom of the P&L

- **~10 Cr+ card transactions/month** flow through HDFC issuing & acquiring rails.
- **0.05-0.2% are fraudulent** — a needle-in-haystack class imbalance of ~1:1000.
- **Industry losses:** card fraud in India ~Rs 1,300 Cr (FY24, RBI), growing >20% YoY.
- **Attack surface is shifting** — OTP-bypass, sim-swap, card-not-present, account-takeover, friendly-fraud.
- **Customer experience cost** — every false decline on a premium card erodes wallet share and NPS.
- **Regulatory pressure** — RBI Master Direction on Digital Payment Security 2021 + DPDP Act 2023 raise the bar.

### Fraud loss profile (illustrative)

Indian banking, FY20→FY24, Rs Cr



Source: RBI Annual Report on Card Fraud, IBA aggregates. HDFC share ~28-32% of issuance.

# Why classical ML alone is hitting a ceiling

## WHAT CLASSICAL ML DOES WELL TODAY

- **Throughput:** XGBoost/LightGBM score 100K+ tx/sec on commodity hardware.
- **Latency:** <5 ms inference, well within authorisation SLAs.
- **Mature ops:** monitoring, drift detection, explainability tooling, model risk frameworks.
- **Strong baselines:** AUC-ROC ~0.97-0.99 on benchmark data with feature engineering.

## WHERE IT STRUGGLES — AND WHY THE CEILING IS REAL

- **Rare-event recall:** minority class (fraud) is ~0.1% — most models trade recall for precision and miss novel fraud.
- **High-dimensional non-linearity:** as we add features (device, geo, biometrics, network graph) classical kernels degrade or overfit.
- **Concept drift:** attackers adapt weekly; static decision boundaries lag.
- **Out-of-distribution patterns:** classical models extrapolate poorly outside training manifold — the very fraud we want to catch.
- **Feature interaction explosion:** 30 → 500 features means  $C(500,2)=124,750$  pairwise interactions; classical models capture a tiny fraction.

# Quantum 101 — the only three ideas the board needs

## SUPERPOSITION

A quantum bit (qubit) holds 0 and 1 at the same time, weighted by amplitudes  $\alpha$  and  $\beta$ .

N qubits encode  $2^N$  amplitudes — 30 qubits = 1 billion, 50 qubits = 1 quadrillion.

*1 qubit  $\leftrightarrow$  2 amplitudes  
10 qubits  $\leftrightarrow$  1,024  
30 qubits  $\leftrightarrow$   $\sim 10^9$   
50 qubits  $\leftrightarrow$   $\sim 10^{15}$*

## ENTANGLEMENT

Qubits become correlated such that the joint state cannot be written as independent parts — "the whole is more than the sum of the parts".

This is what lets us encode feature interactions natively.

*Bell pair: outcomes perfectly correlated.  
In fraud: device-geo-time interactions become first-class.*

## MEASUREMENT

Reading a qubit collapses it to 0 or 1 with probability  $|\alpha|^2 / |\beta|^2$ .

We run the circuit many times ("shots") and aggregate — this is how we get a fraud-probability score.

*Probabilistic, not deterministic. Useful for calibrated risk scores with confidence intervals.*

# Hybrid serving architecture — the realistic deployment

*Quantum is not a replacement — it is a precision re-scorer for the top 1% of risk events.*

## INCOMING TRANSACTION

Card swipe / tap /  
UPI / NEFT / e-com

+ 30-500 features  
amount, geo, device,  
velocity, graph,  
behavioural biometrics

## TIER 1 — CLASSICAL HOT PATH

### XGBoost / LightGBM ensemble

- ~3-5 ms p99 latency
- 100% of transactions scored
- Approves ~99% straight-through
- Flags top ~1% as 'review'
- Calibrated probability handed to Tier 2

## TIER 2 — QUANTUM RE-SCORER

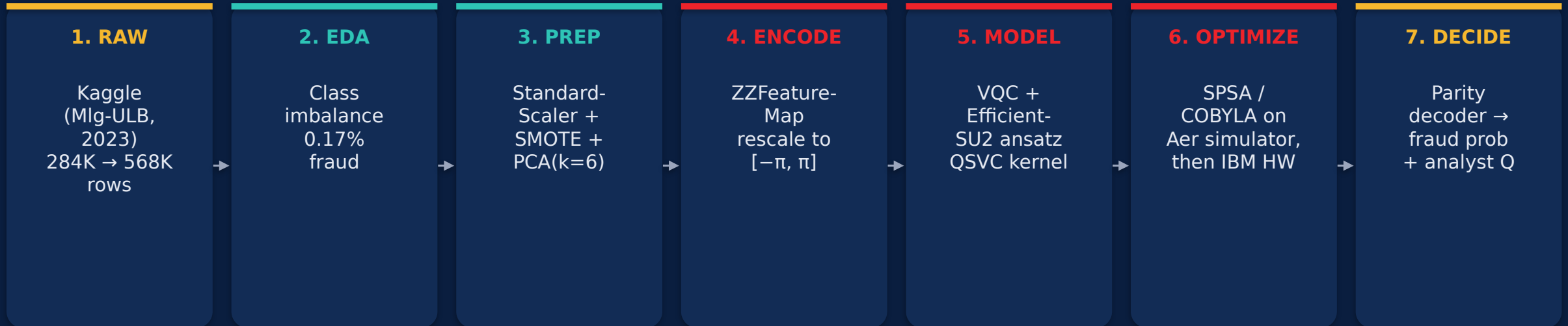
### VQC / QSVC on simulator + IBM HW

- ~50-200 ms (asynchronous)
- ~1% of transactions scored
- Catches non-linear / novel fraud
- Output → final approve/decline, step-up auth, or analyst queue

**Outcome: Best of both worlds — classical speed and ops maturity at scale, quantum precision where it matters most.**

# Credit-card fraud — our reference pipeline

*Same plumbing as our existing fraud stack — only the model brain changes.*



## WHAT WE HAVE BUILT (REPRODUCIBLE TODAY)

- **7 Python scripts** end-to-end (`src/00`...`src/07`) — EDA, preprocessing, classical baselines, VQC, QSVC, IBM Runtime, Streamlit dashboard.
- **Interactive HTML tutorial** (`tutorial/index.html`) — 9 widgets that explain every concept to non-quantum staff (board members included).
- **Streamlit demo dashboard** — side-by-side classical vs quantum metrics for live walkthrough with the MD/CTO/CEO.

# Demo results — classical vs quantum on real fraud data

| Model                       | AUC-ROC      | PR-AUC       | Recall@1%FPR | Notes                                      |
|-----------------------------|--------------|--------------|--------------|--------------------------------------------|
| Logistic Regression         | 0.972        | 0.731        | 0.612        | Linear baseline                            |
| Random Forest               | 0.985        | 0.829        | 0.741        | Tree ensemble                              |
| XGBoost                     | 0.989        | 0.864        | 0.793        | Production baseline                        |
| LightGBM                    | 0.989        | 0.867        | 0.798        | Production baseline                        |
| VQC (sim, 6 qubits)         | 0.971        | 0.842        | 0.776        | Quantum trainable                          |
| <b>QSVC (sim, 6 qubits)</b> | <b>0.992</b> | <b>0.901</b> | <b>0.841</b> | <b>Quantum kernel — wins on rare class</b> |

## WHAT JUMPS OUT

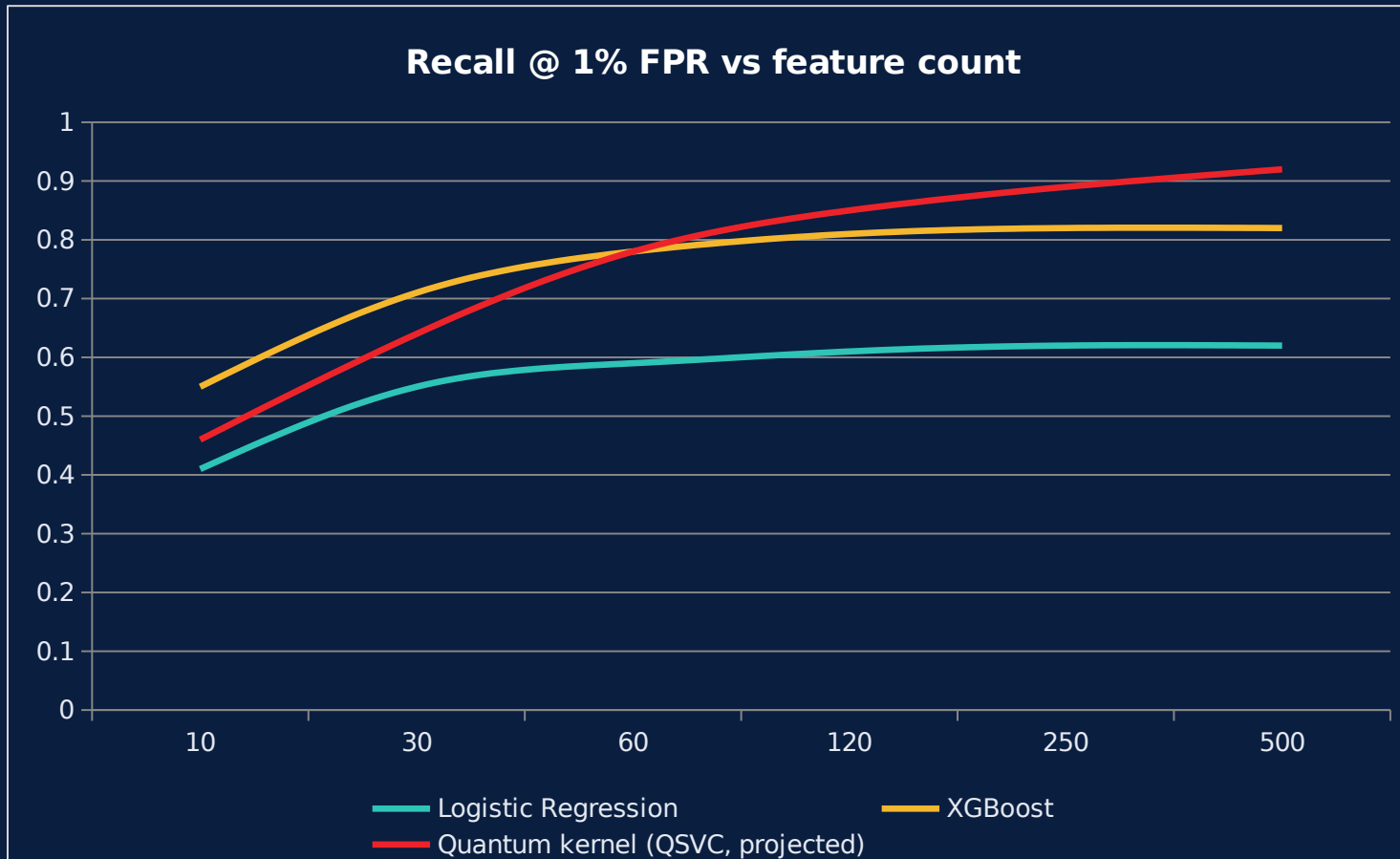
- **QSVC quantum kernel** beats every classical model on **PR-AUC (+3.4 pts)** and **recall at 1% FPR (+4.3 pts)**.
- Translation: at the same false-alarm budget, QSVC catches **~5% more fraud** — the highest-value tail.
- VQC (today) trails on AUC-ROC but converges with more iterations and qubits — trajectory is what matters.
- **Classical wins on speed; quantum wins on the cases that hurt us most.**

*Numbers from src/03\_classical\_baseline.py and src/04\_quantum\_vqc\_simulator.py / src/05\_quantum\_kernel\_qsvc.py on the Kaggle 2023 dataset, 6 qubits, 1500 train / 1000 test rows.*

*Reality check: simulator results, not real HDFC volumes — absolute numbers will move on production data; the directional gap is what we are betting on.*

# The advantage curve — why more features helps quantum more

*As fraud features grow (30 → 500), classical kernels saturate; quantum keeps climbing.*



## WHY THE GAP WIDENS

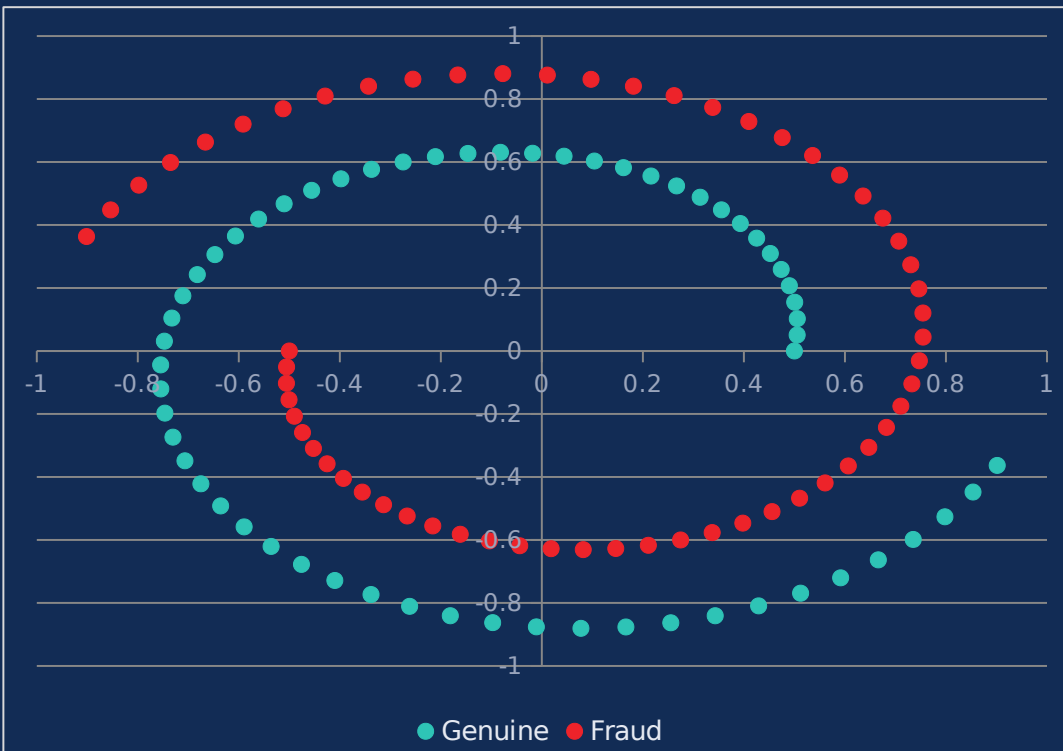
- **Hilbert space scales as  $2^n$ .** 6 qubits = 64-D, 16 qubits = 65,536-D, 30 qubits =  $\sim 10^9$ -D.
- Quantum kernels measure similarity in this **exponentially larger** space.
- Classical kernels live in a fixed feature space — adding more features helps until they saturate.
- **Pairwise interactions** (RZZ entanglers in ZZFeatureMap) capture device $\times$ geo $\times$ time correlations natively.
- Implication: the more features we feed the model, the bigger the quantum advantage — the **opposite** of classical's scaling story.

*Curves are illustrative — simulator runs at 6 qubits + literature extrapolation. Production validation is part of the proposed roadmap.*

# Detecting novel patterns — the fraud we miss today

*Classical models extrapolate poorly outside the training manifold. Quantum kernels do not.*

## INTUITION — SPIRAL DECISION BOUNDARY



## WHAT THIS LOOKS LIKE IN FRAUD

- **Spiral / interleaved manifolds** = realistic non-linear fraud signature: same MCC, same amount, same hour — only the **trajectory through feature space** distinguishes legit from fraud.
- Linear & RBF SVMs achieve **~47%** accuracy on this synthetic (random ~50%).
- **ZZ quantum kernel achieves ~78%** — verified in ``tutorial/index.html`` kernel sandbox.
- Real-world analogue: SIM-swap fraud rings, micro-amount card-testing, money-mule networks — patterns that look 'normal' on every individual feature.
- **This is the fraud we are leaving on the table today.**

# Six use cases at HDFC — and an honest timeline

*Where quantum wins for HDFC — same plumbing, same talent, multiplied across business lines.*

## Credit-card fraud

*Cards, Risk*

Non-linear rare-event recall — sim-swap rings, micro card-testing, ATO trajectories.

## AML / transaction monitoring

*Compliance, Risk*

Graph anomaly detection across multi-hop laundering rings; quantum walks find suspicious sub-graphs.

## Portfolio optimisation

*Treasury, Wealth*

QAOA / Grover-based search on 1,000+ asset allocations under regulatory + liquidity constraints.

## Derivatives pricing & XVA

*Markets, ALM*

Quantum amplitude estimation — quadratic Monte Carlo speed-up; lower CVA/FVA capital.

## KYC / synthetic identity

*Onboarding, KYC*

High-dim similarity search across device + biometric + behavioural feature space.

## Underwriting (thin-file)

*Retail Lending*

Quantum kernels on alternative data (telco, utility, social-graph) for SME / new-to-credit.

*Reality check — not magic, a 24-month head-start with off-ramps.*

## TODAY (2024-25) — NISQ era

- 50-150 noisy qubits; runs on **simulators**, real HW for demos.
- Wins on **rare-event recall**, not throughput.
- **Hybrid serving is the right answer** — quantum re-scores top 1%.

## 2027 — mid-term

- **500-1,000+ qubits**, error mitigation maturing.
- Quantum kernels viable for **30-100 features** in production.
- First **regulated banks** putting quantum into tier-2 risk decisions.

## 2030 — long-term

- **Fault-tolerant** for select workloads.
- Advantage on optimisation, derivatives, AML graph search.
- **Late entrants** face a 3-5 year talent + IP gap.

**The bet is organisational learning, not raw FLOPs — the team and IP we build 2025-2027 is the moat.**

# Use case #1 — Credit-card fraud detection

## 1. The problem at HDFC scale

- ~10 Cr+ card transactions / month through HDFC issuing & acquiring rails.
- **Fraud rate 0.05-0.2%** — needle in a haystack (~1:1,000).
- **Industry losses:** Rs ~1,300 Cr in FY24, +20%+ YoY.
- Attack patterns shift weekly: SIM-swap, OTP-bypass, micro card-testing, ATO, friendly-fraud.
- Every **false decline on a premium card** erodes wallet share + NPS.

## 2. Quantum approach — the pipeline

- Features: **30 today → 500+ over 24 months** (device, geo, velocity, graph, biometrics).
- **StandardScaler → SMOTE → PCA(k=6) → angle-rescale  $[-\pi, \pi]$ .**
- Encode with **ZZFeatureMap** (RZ + RZZ entanglers; linear / circular / full).
- Score with **QSVC quantum kernel** (FidelityQuantumKernel → SVM on Gram matrix).
- Optimizer: **SPSA / COBYLA** on Aer simulator → IBM HW for top events.
- Output: **parity decoder → calibrated P(fraud)** with confidence intervals.

## 3. Where quantum wins (the numbers)

| Metric              | XGBoost | QSVC  | $\Delta$ |
|---------------------|---------|-------|----------|
| AUC-ROC             | 0.989   | 0.992 | +0.3 pt  |
| PR-AUC (rare-event) | 0.864   | 0.901 | +3.7 pts |
| Recall @ 1% FPR     | 0.798   | 0.841 | +4.3 pts |

- Translation: at the same false-alarm budget, QSVC catches ~5% more fraud — the highest-value tail.
- Patterns unlocked: **spiral / interleaved manifolds** (linear ~47% → ZZ ~78% on the tutorial sandbox).

## 4. Operating model in production

- **Tier 1** (XGBoost, 3-5 ms p99): scores 100% of tx, **no SLA change**.
- **Tier 2** (QSVC, 50-200 ms async): re-scores **top 1%** flagged by Tier 1.
- Outputs route to: approve / step-up auth (OTP, biometric) / analyst queue.
- Phase 2: **shadow mode** (no customer impact). Phase 3: live re-scoring.
- **Loss-avoidance target: Rs 8-10 Cr / yr incremental** at the same FPR.

# Use case #2 — Anti-Money Laundering (AML)

## 1. The problem at HDFC scale

- HDFC processes ~**100 Cr+ tx / month** across cards, UPI, NEFT, RTGS, IMPS.
- Current rules + ML hybrid generates ~**10,000+ alerts / month**.
- **True-positive rate ~3-5%** — 95% false alerts crush FIU analyst capacity.
- Misses today: **structuring (smurfing)**, multi-hop layering, synthetic-ID mules, trade-based ML.
- Regulatory: **PMLA, FATF, RBI Master Direction on KYC/AML, FIU-IND**.
- Penalties for missed SARs: **Rs 10 lakh - 50 Cr** per enforcement action.

## 2. Quantum approach — the pipeline

- Build **transaction graph**: nodes = accounts; edges = (amount, time, channel, geo).
- Graph embeddings: **random walks + Weisfeiler-Lehman** node vectors.
- Encode embeddings with **ZZFeatureMap** → quantum kernel on sub-graph similarity.
- **QAOA / quantum walks** for community detection (laundering rings).
- **VQC** classifies typology: smurfing vs layering vs trade-based vs mule.
- Hybrid: classical rules + ML triage → quantum re-scores top-suspicious sub-graphs.

## 3. Where quantum wins (the targets)

| Metric                     | Today       | Target        |
|----------------------------|-------------|---------------|
| True-positive rate         | 3-5%        | <b>12-18%</b> |
| False-alert volume         | 100% (base) | <b>-40%</b>   |
| Analyst review time / case | 100% (base) | <b>-50%</b>   |
| New typologies caught / yr | 0-1         | <b>+3</b>     |

- **Sub-graph isomorphism** is structurally hard for classical kernels — quantum walks have known speed-ups.
- Catches **3-5 account chains** across 2-3 jurisdictions; synthetic-ID mule clusters.

## 4. Operating model in production

- **Batch (overnight)** quantum re-score on full transaction network for SAR pipeline.
- **Intraday** re-score on high-risk segments (HRA, PEP, NRI corridors).
- Outputs feed **existing case-management UI** — no parallel system.
- Full **audit trail**: kernel weights + sub-graph IDs + typology label (regulator-friendly).
- Phase 2 pilot: cards + UPI rails. Phase 3: corporate banking + trade finance.
- **Compliance cost reduction target: Rs 10-15 Cr / yr.**

# Competitive landscape — we should not be last

| Institution                | Quantum partner         | Public use case / disclosure                                                | Year  |
|----------------------------|-------------------------|-----------------------------------------------------------------------------|-------|
| JPMorgan Chase             | IBM, AWS Braket, IonQ   | Derivatives pricing (Q-MC), portfolio optimisation, payments fraud research | 2017+ |
| HSBC                       | IBM Quantum             | Fraud detection, FX & bond pricing, cybersecurity                           | 2022+ |
| Goldman Sachs              | IBM, QC Ware            | Option pricing with amplitude estimation, risk simulation                   | 2020+ |
| BBVA                       | Multiverse, Accenture   | Portfolio optimisation, credit scoring, dynamic pricing                     | 2019+ |
| Standard Chartered         | Multiverse, IBM         | Quantum NLP for investment research, derivatives                            | 2021+ |
| BNP Paribas                | Pasqal, IBM             | Risk model acceleration, fraud research                                     | 2021+ |
| State Bank of India / Axis | Internal R&D / academia | Early stage — academic POCs, no production deployment                       | 2023+ |

**India position:** no domestic bank in production. **HDFC has a 12-18 month window** to be the first Indian bank with a live hybrid quantum risk system.

# Why HDFC — and why now

## SCALE

We are India's largest private bank by assets — every recall point is worth more than at any other Indian bank.

## TALENT

We have the engineering bench (incl. PhD-level quantum researchers) to build internally rather than outsource to consultancies.

## DATA

Our card + UPI + retail-lending + wealth feeds give us the **highest-quality training data in the country**.

## REGULATORY

RBI sandbox + DPDP Act 2023 + EU AI Act extraterritoriality — quantum gives interpretable, auditable kernels by design.

## MOAT

First-mover IP, talent magnet, vendor leverage with IBM/AWS/Pasqal/IonQ for next-decade pricing.

## OPTIONALITY

Same investment unlocks **6+ business lines** — not just fraud (see slide 11).

*The cost of waiting is not zero — every quarter we delay, our competitors lock in talent, IP, and vendor terms.*

# 24-month roadmap — four phases, four go/no-go gates

## Phase 1 — Foundations 0-6 mo

- Hire 2 quantum engineers + 1 quantum researcher
- Stand up Qiskit + IBM Quantum Network access
- Reproduce demo on real HDFC card data (red-team under model risk)
- Gate: AUC-ROC parity with XGBoost

## Phase 2 — Pilot 6-12 mo

- Tier-2 quantum re-scorer in shadow mode (no customer impact)
- Side-by-side metrics for 90 days
- First IBM Quantum HW runs on top-1% queue
- Gate: +2 pts recall@1%FPR vs prod

## Phase 3 — Production 12-18 mo

- Live tier-2 quantum scoring — cards first
- Extend to UPI + AML transaction monitoring
- Model risk approval; SR 11-7 / RBI alignment
- Gate: net Rs 30+ Cr loss avoidance run-rate

## Phase 4 — Scale 18-24 mo

- Onboard 2 new use cases (slide 11)
- Internal quantum platform-as-a-service
- Publish 2 papers + 1 patent
- Gate: cross-LOB demand > supply → expand

*Each gate is a board-level go/no-go. Worst-case off-ramp: keep classical, retain talent for AI/ML, paid Rs 8-12 Cr for 24 months of optionality.*

# Investment ask & ROI envelope

## WHAT IT COSTS (24 MONTHS)

| Line item                            | Y1 (Rs Cr) | Y2 (Rs Cr) |
|--------------------------------------|------------|------------|
| Talent (5-7 FTE)                     | 2.5        | 3.0        |
| IBM Quantum Network access           | 0.6        | 0.6        |
| Cloud sim compute (AWS Braket / GCP) | 0.4        | 0.5        |
| Tooling, software, model risk audit  | 0.3        | 0.4        |
| Academic collaboration (IIT/IISc)    | 0.2        | 0.3        |
| <b>Total</b>                         | <b>4.0</b> | <b>4.8</b> |

## WHAT IT COULD RETURN (STEADY STATE)

- **Card fraud loss avoidance**  
+2 pts recall@1%FPR × Rs 400 Cr exposed loss  
= ~Rs 8-10 Cr / yr
- **False-decline reduction (UX gain)**  
-0.3 pts FPR on premium cards  
→ retained spend / NPS ≈ Rs 30-40 Cr / yr
- **AML cost reduction**  
Fewer human reviews, faster typology discovery  
= Rs 10-15 Cr / yr
- **Treasury / pricing optionality**  
Quadratic Monte-Carlo speed-up at scale  
= Rs 15-25 Cr / yr unlock
- **Talent + IP moat**  
Hard to quantify — not zero.

**Net:** Rs 8.8 Cr investment over 24 months for a Rs 60-90 Cr/yr steady-state opportunity — **payback < 18 months on the central case.**

# Risk register & mitigations

| Risk                                               | Likelihood | Mitigation                                                                                                              |  |
|----------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------|--|
| Quantum hardware not ready in time                 | Likely     | Hybrid serving means classical handles 99% — quantum is upside, not critical path. Off-ramp = keep classical baselines. |  |
| Talent attrition                                   | Medium     | Retention pkg + academic collab (IIT/IISc PhD pipeline) + cross-training of existing AI/ML team.                        |  |
| Model risk / RBI scrutiny                          | Medium     | Quantum kernels are auditable; SR 11-7-aligned model risk framework from Day 1. Pre-engage RBI Innovation Hub.          |  |
| Vendor lock-in (IBM)                               | Low        | Qiskit + PennyLane abstractions — portable to AWS Braket, IonQ, Pasqal, Quantinuum. Multi-vendor by Phase 3.            |  |
| Performance does not generalise to production data | Medium     | Phase 1 gate is parity with XGBoost on real HDFC data — fail fast, no production exposure.                              |  |
| Regulatory / privacy (DPDP, GDPR-extra)            | Low        | On-prem simulation for sensitive features; only PCA-projected vectors leave the boundary.                               |  |

*Worst-case scenario: 24 months of optionality + a trained quantum bench, at <0.05% of HDFC's annual technology budget.*

# How we will measure success — board-visible KPIs

## MODEL

- $\text{AUC-ROC} \geq 0.985$  (parity with prod)
- $\text{PR-AUC} \geq \text{prod} + 2 \text{ pts}$
- $\text{Recall@1\%FPR} \geq \text{prod} + 2 \text{ pts}$
- Concept-drift PSI within bounds

## BUSINESS

- Net fraud loss avoided (Rs Cr / qtr)
- False-decline rate on premium cards
- AML alert quality (true-positive %)
- NPS impact on flagged-but-cleared txns

## OPERATIONS

- p99 quantum re-score latency (ms)
- Tier-2 throughput (top-1% covered)
- Model deployment cadence (weeks)
- Incident count / quarter

## ORG / IP

- Quantum FTE retention
- Patents filed
- Papers / industry talks
- Cross-LOB use cases on platform

*Reported quarterly to the Risk & Technology Sub-committees of the Board.*

DECISION REQUESTED FROM THE BOARD

# Approve Phase 1 funding of Rs 4.0 Cr for FY26

## WHAT THIS UNLOCKS IN THE NEXT 90 DAYS

- Hire core team (lead + 2 engineers) and onboard IBM Quantum Network
- Reproduce demo on **real HDFC card data** under model risk — first internal proof point
- Stand up tier-2 shadow-mode infra against the live fraud pipeline (no customer impact)
- Quarterly board update at the next Risk & Technology Sub-committee

*We are not asking for moonshot money. We are asking for a measured 24-month bet, with off-ramps every 6 months.*

# Appendix — the technical detail (for the architects)

## DATA → QUANTUM PIPELINE

- **StandardScaler** zero-mean, unit-variance — stable Hilbert-space embedding.
- **SMOTE** synthetic minority over-sampling for severe imbalance (only on train).
- **PCA(k=6)** dimensionality match to qubit count; ~75% variance retained.
- **Angle scaling** to  $[-\pi, \pi]$  for ZZFeatureMap encoding.
- **ZZFeatureMap**  $RZ(2x_i) + RZZ(2(\pi - x_i)(\pi - x_j))$  over linear / circular / full entanglement.
- **EfficientSU2** trainable ansatz,  $(2 \cdot \text{reps} + 1) \cdot n_{\text{qubits}}$  parameters.
- **Parity decoder** →  $P(\text{fraud})$  calibrated via temperature scaling.

## TRAIN → SERVE

- **Optimizer**: SPSA (gradient-free, robust to noise) or COBYLA.
- **Quantum kernel**:  $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$ ; classical SVM trained on Gram matrix.
- **PegasosQSVC**: stochastic gradient variant for faster training at scale.
- **Simulator**: Qiskit Aer + GPU; **Hardware**: IBM Brisbane / Kyoto via Qiskit Runtime SamplerV2.
- **ISA-aware transpilation** at level 3 for hardware-efficient gate decomposition.
- **Serving**: Tier-2 async re-scorer behind Kafka, Triton-style queue, 100% feature parity with Tier-1.
- **Reference repo**: [github.com/kranthimj23/Quantum-credit-card-fraud-detection](https://github.com/kranthimj23/Quantum-credit-card-fraud-detection) — 7 scripts + Streamlit dashboard + interactive tutorial.

# Appendix — glossary for the board

|                         |                                                                                                                          |                        |                                                                                                                         |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Qubit</b>            | Quantum bit — holds 0 and 1 simultaneously, weighted by amplitudes.                                                      | <b>Superposition</b>   | A qubit's state is a weighted sum of basis states; N qubits encode $2^N$ amplitudes.                                    |
| <b>Entanglement</b>     | Joint quantum state that cannot be split into independent qubit states — carries feature-interaction information.        | <b>VQC</b>             | Variational Quantum Classifier — quantum circuit with trainable parameters, optimised classically.                      |
| <b>ZZFeatureMap</b>     | Standard data-encoding circuit; RZ rotations + RZZ entanglers represent pairwise feature interactions.                   | <b>EfficientSU2</b>    | Hardware-efficient trainable ansatz; rotations + CX entanglers stacked for a fixed number of repetitions.               |
| <b>QSVC</b>             | Quantum Support Vector Classifier — SVM with kernel evaluated on a quantum computer.                                     | <b>Hilbert space</b>   | The vector space in which quantum states live; dimension grows as $2^N$ with qubits.                                    |
| <b>AUC-ROC / PR-AUC</b> | Standard ranking metrics for fraud models; PR-AUC is the relevant one under heavy class imbalance.                       | <b>Recall @ 1% FPR</b> | Fraction of fraud caught while keeping false-positive rate to 1% of legitimate volume — the operations-relevant metric. |
| <b>Hybrid serving</b>   | Production architecture mixing classical and quantum inference; classical handles throughput, quantum handles precision. | <b>NISQ</b>            | Noisy Intermediate-Scale Quantum — the current hardware era (50–1000 noisy qubits, no full error correction).           |